# 🩺 Public Health Analytics Dashboard

## 📊 Diabetes predictor- Patient Health Analysis

| 👥 Total Patients | 🍬 Average Sugar | ⚠️ High Risk |
|---|---|---|
| 15 | 140.33 | 8 |

## 📋 Patient Records

| | ↑ Age | BMI | BP | Sugar | O |
|---|---|---|---|---|---|
| 10 | 38 | 30 | 89 | 145 | |
| 6 | 40 | 27 | 86 | 130 | |
| 14 | 41 | 29 | 87 | 135 | |
| 2 | 45 | None | 90 | 150 | |
| 11 | 48 | 32 | 92 | 155 | |
| 3 | 50 | 29 | 88 | 140 | |
| 12 | 52 | 34 | 94 | 170 | |
| 7 | 55 | 33 | 91 | 160 | |
| 4 | 60 | 35 | 95 | 180 | |
| 8 | 65 | 36 | 96 | 190 | |

## 🧹 Missing Values Handling

| | 0 |
|---|---|
| Age | |
| BMI | |
| BP | |
| Sugar | |
| Outcome | |

Missing values filled using median

## 📊 Descriptive Statistics

| | Age | BMI | BP | Sugar |
|---|---|---|---|---|
| count | 15 | 15 | 15 | 15 |
| mean | 43 | 29.1333 | 88.4333 | 140.3333 |
| std | 11.9403 | 4.3072 | 4.7767 | 27.5465 |
| min | 25 | 22 | 80 | 100 |
| 25% | 34 | 26.5 | 85.5 | 117.5 |
| 50% | 41 | 29 | 88.5 | 140 |
| 75% | 51 | 32.5 | 91.5 | 157.5 |
| max | 65 | 36 | 96 | 190 |

## 📌 Mean Values

| | 0 |
|---|---|
| Age | |
| BMI | |
| BP | |
| Sugar | |
| Outcome | |

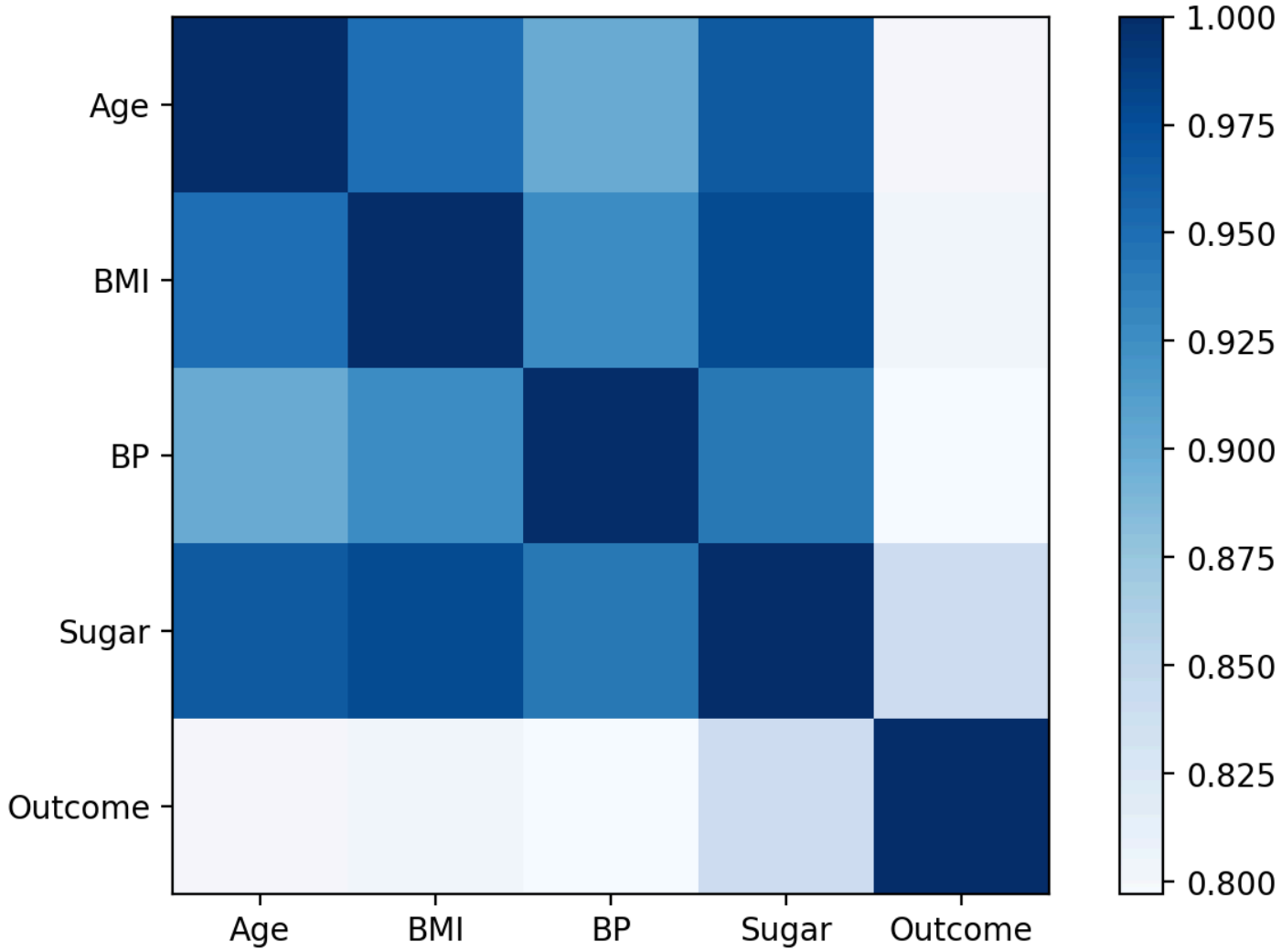
## 📌 Standard Deviation

| | 0 |
|---|---|
| Age | |
| BMI | |
| BP | |
| Sugar | |
| Outcome | |

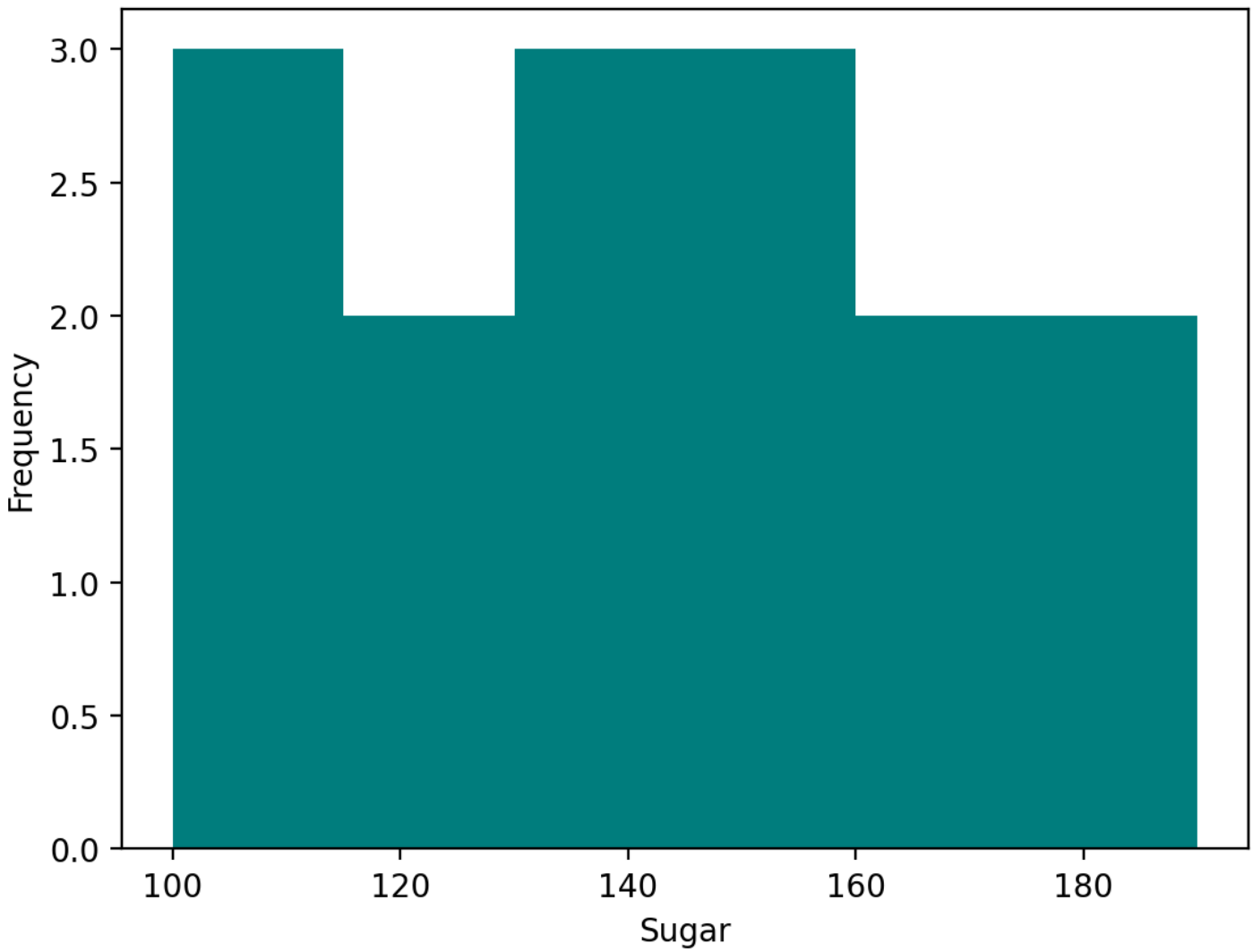
## 📏 Data Normalization

Normalization completed

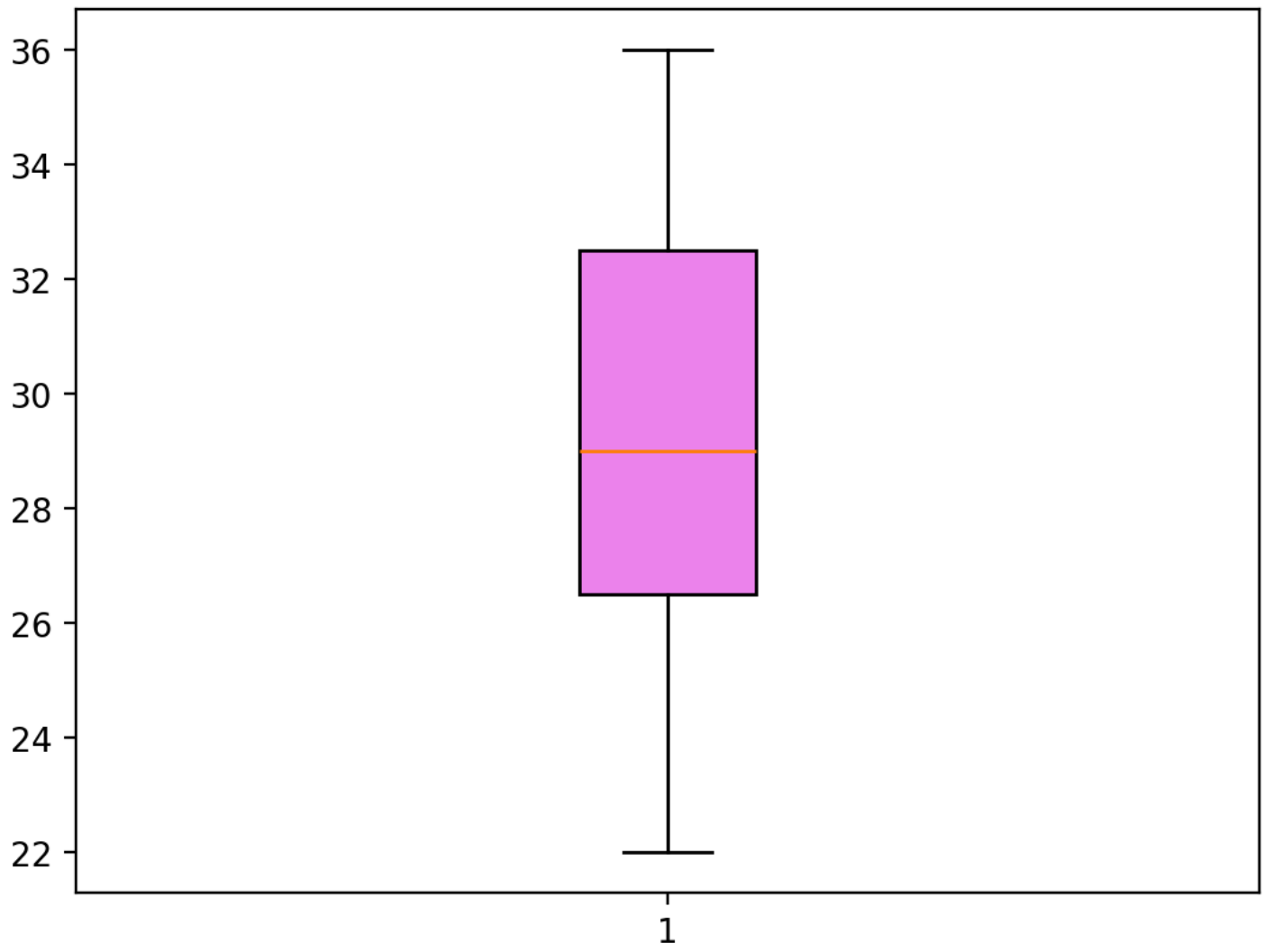
## 🔥 Correlation Heatmap



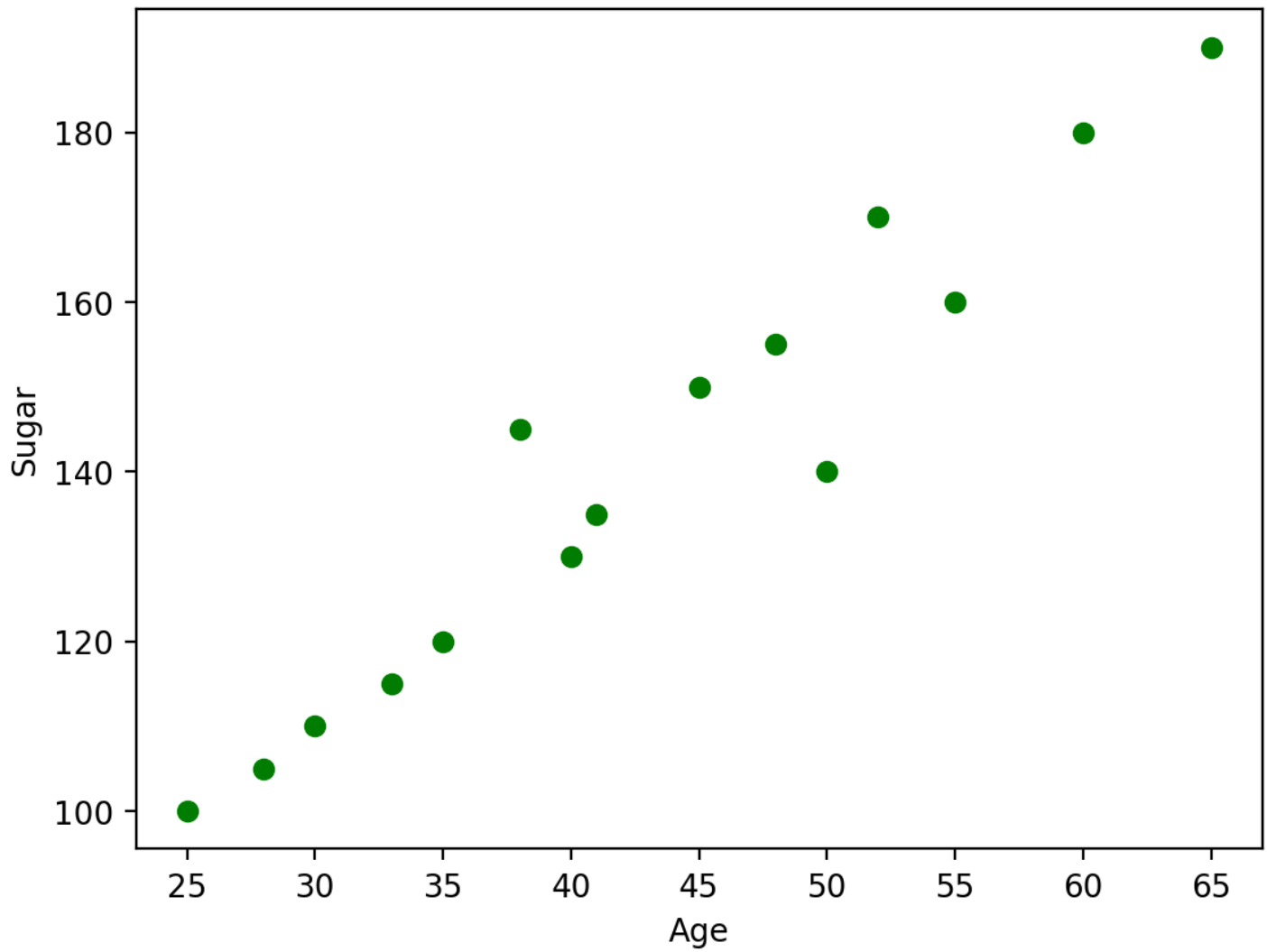

📈 **Sugar Histogram**




📦 BMI Box Plot




📉 Age vs Sugar




## 🎯 Model Accuracy

Accuracy: 100.00%

## 🧪 Patient Risk Prediction

Enter Age

30 − +

Enter BMI

48 − +

Enter BP

99 − +

Enter Sugar

100 − +

🔍 Predict

⚠️ High Diabetes Risk